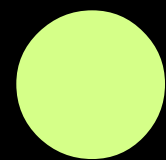


AnomalyDINO: **Boosting Patch-based** **Few-shot Anomaly** **Detection with DINOv2**



Team TensorMind



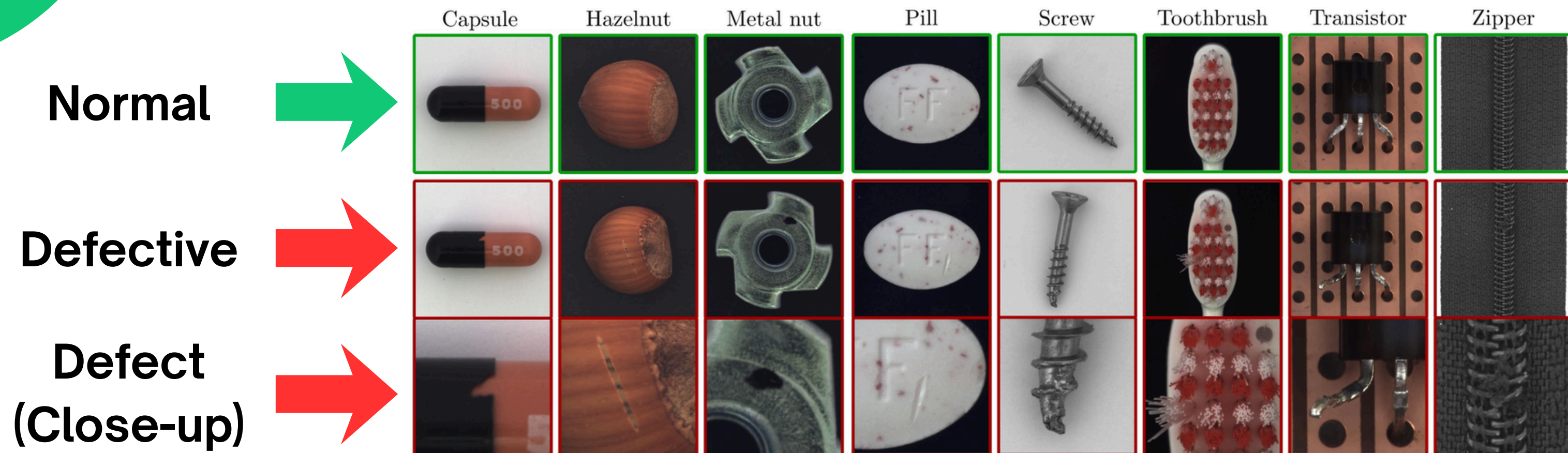
Few-Shot Visual Anomaly Detection

Goal

Detect defects using only a few normal examples

Challenge

Learn what "normal" looks like from very limited data



Research Gap

Existing Limitations

Supervised Models:

Require many labeled anomaly samples

Fine Tuning:

Requires heavy training for each new task

Vision Language:

Powerful, but computationally complex

The Core Question

Few Normal Samples



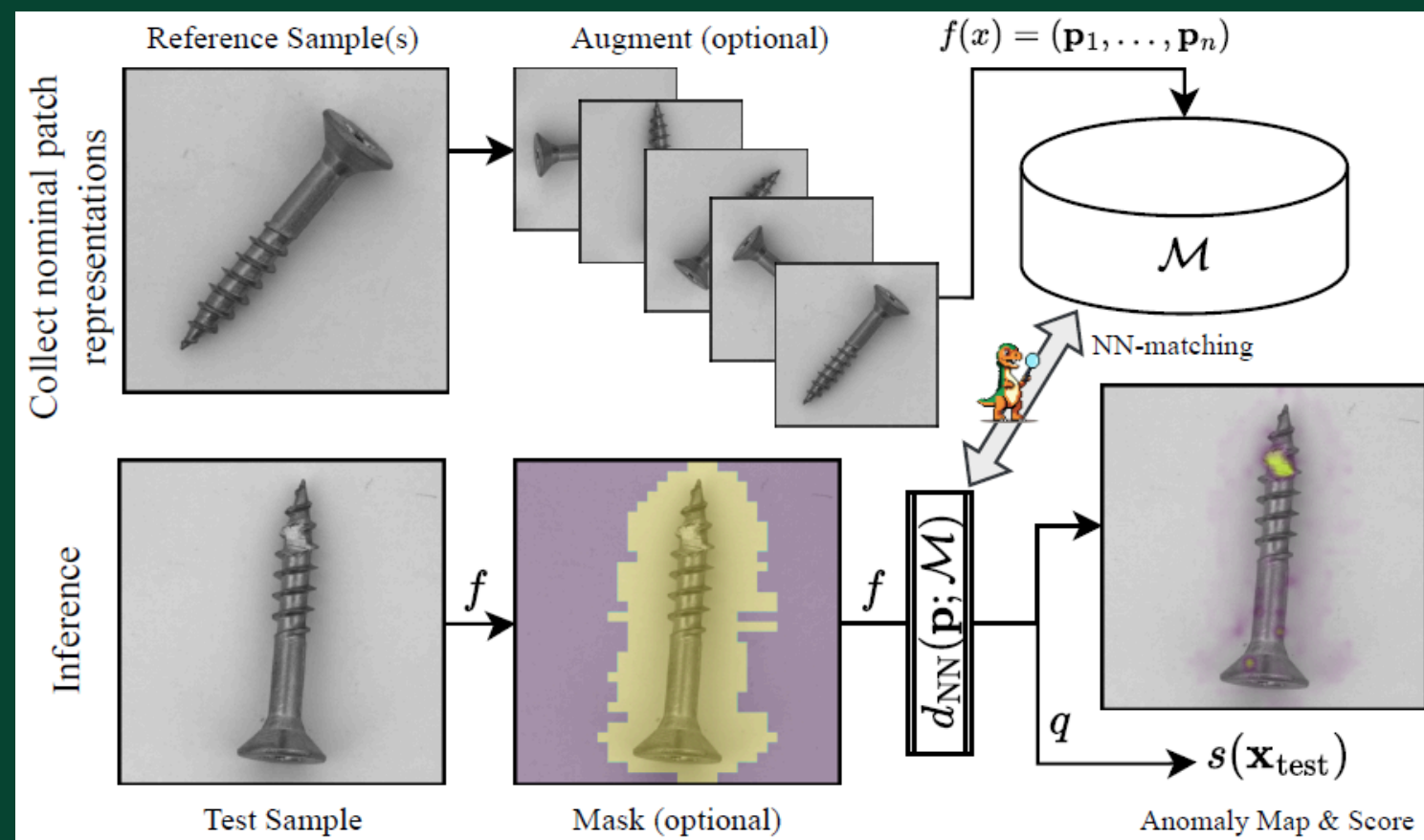
Pretrained Visual Features



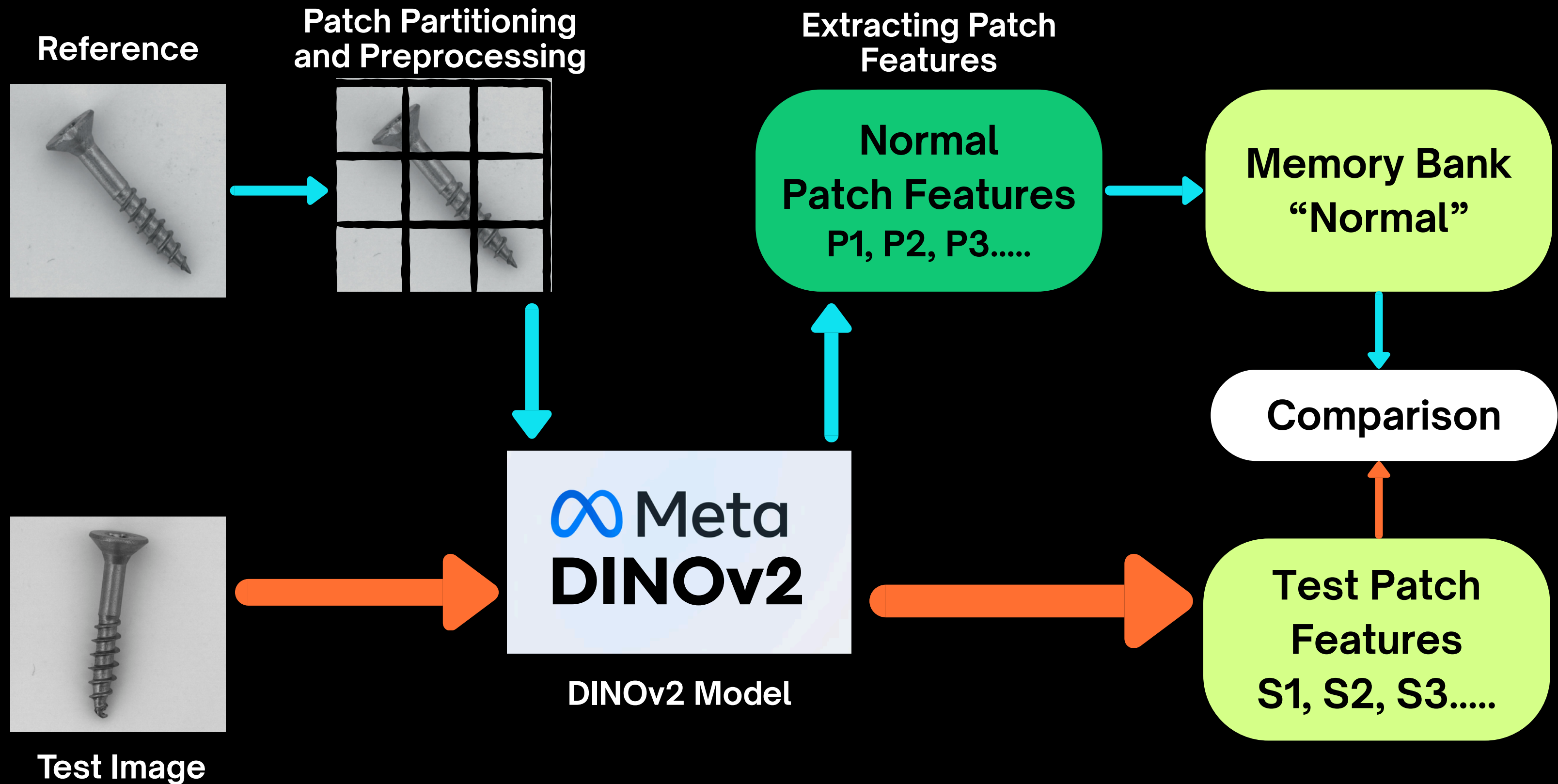
Accurate anomaly detection
without training a new model?



AnomalyDINO

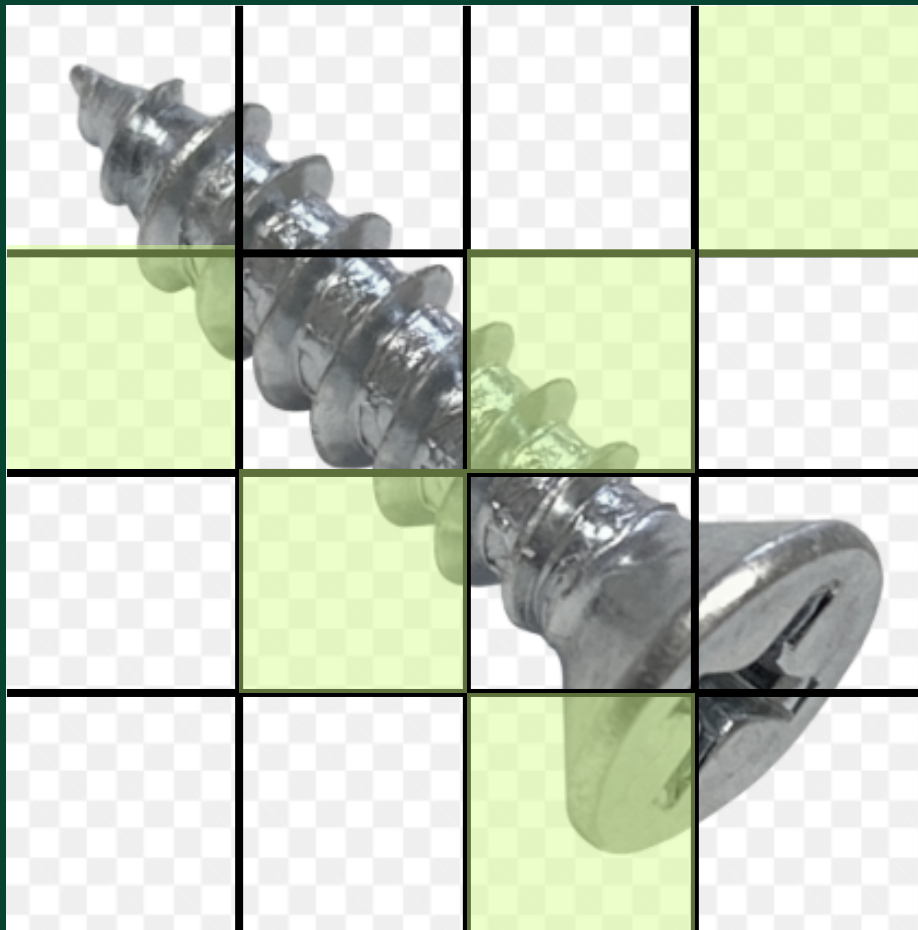


Core Idea of AnomalyDINO



HOW ANOMALYDINO DETECTS THE DEFECT

INPUT IMAGE



DIVIDED INTO 14×14 PATCHES

MEMORY BANK



PATCH-LEVEL DISTANCES

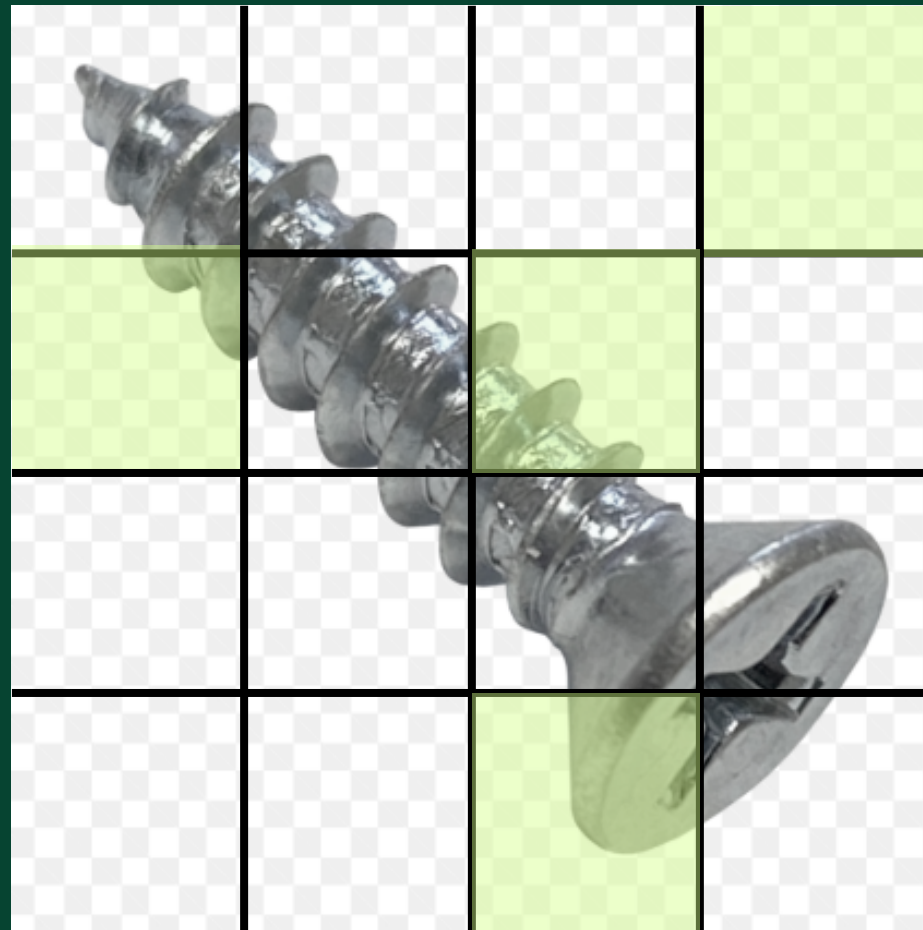
CALCULATIONS

$$d(\mathbf{x}, \mathbf{y}) := 1 - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

$$d_{\text{NN}}(\mathbf{p}; \mathcal{M}) := \min_{\mathbf{p}_{\text{ref}} \in \mathcal{M}} d(\mathbf{p}, \mathbf{p}_{\text{ref}})$$

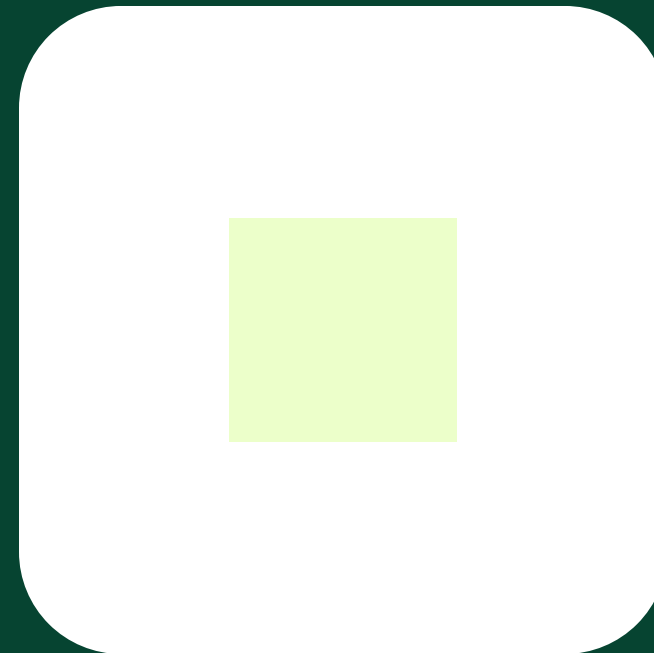
HOW ANOMALYDINO DETECTS THE DEFECT

INPUT IMAGE



DIVIDED INTO 14×14 PATCHES

MEMORY BANK



PATCH-LEVEL DISTANCES

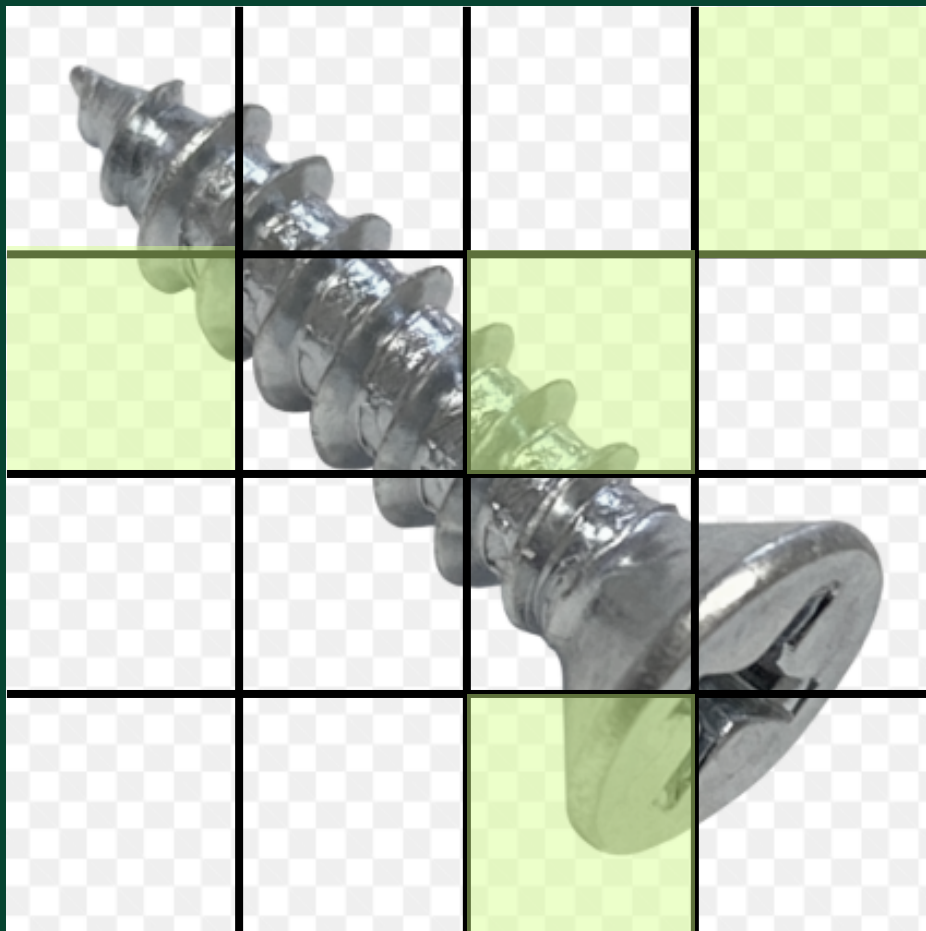
CALCULATIONS

$$d(\mathbf{x}, \mathbf{y}) := 1 - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

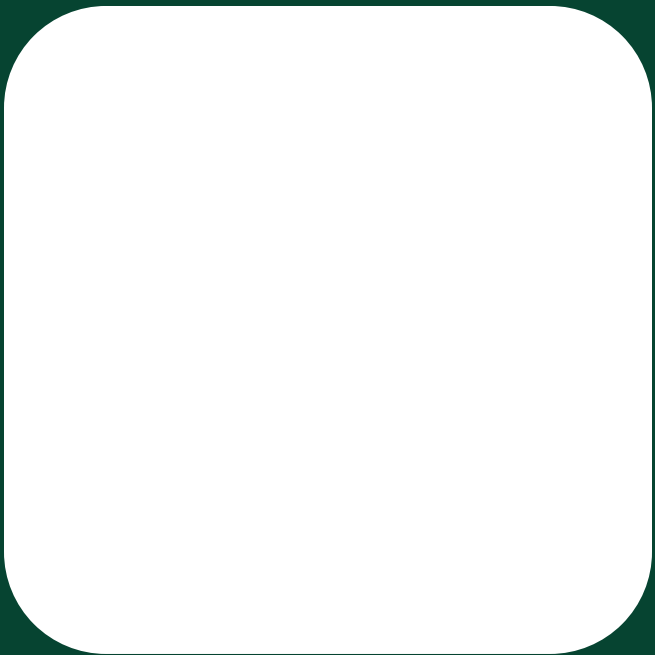
$$d_{\text{NN}}(\mathbf{p}; \mathcal{M}) := \min_{\mathbf{p}_{\text{ref}} \in \mathcal{M}} d(\mathbf{p}, \mathbf{p}_{\text{ref}})$$

HOW ANOMALYDINO DETECTS THE DEFECT

INPUT IMAGE



MEMORY BANK



PATCH-LEVEL DISTANCES



0.82

DIVIDED INTO 14×14 PATCHES

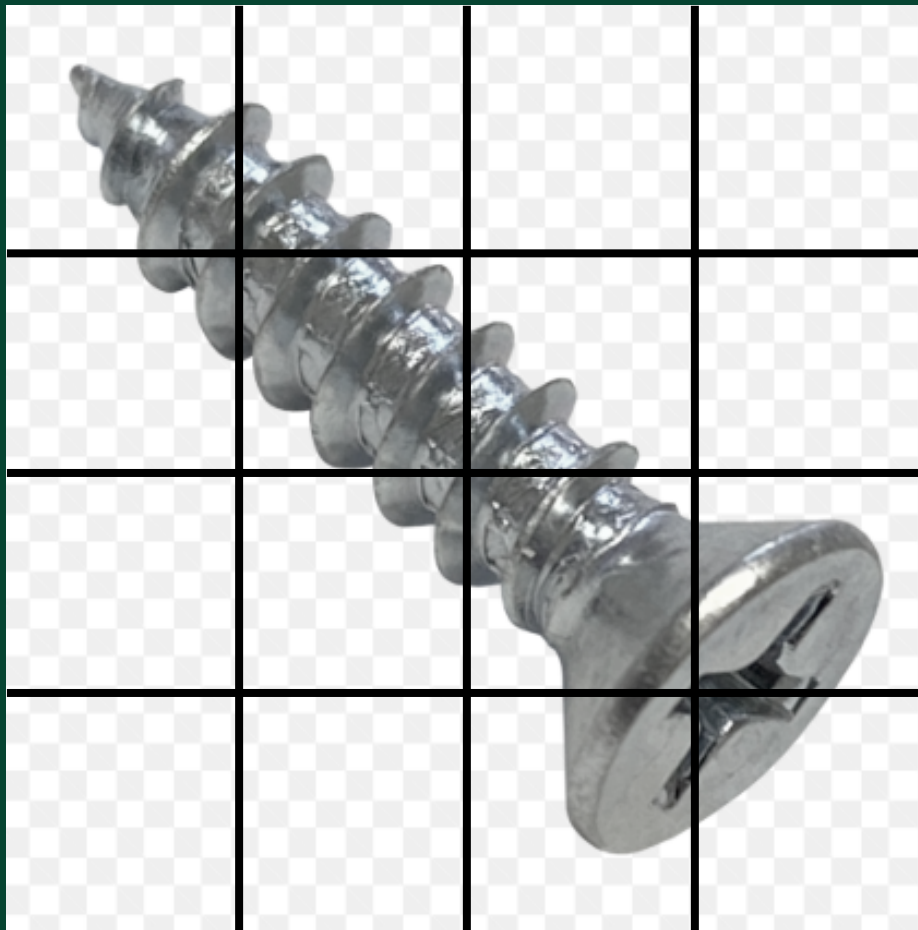
CALCULATIONS

$$d(\mathbf{x}, \mathbf{y}) := 1 - \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

$$d_{\text{NN}}(\mathbf{p}; \mathcal{M}) := \min_{\mathbf{p}_{\text{ref}} \in \mathcal{M}} d(\mathbf{p}, \mathbf{p}_{\text{ref}})$$

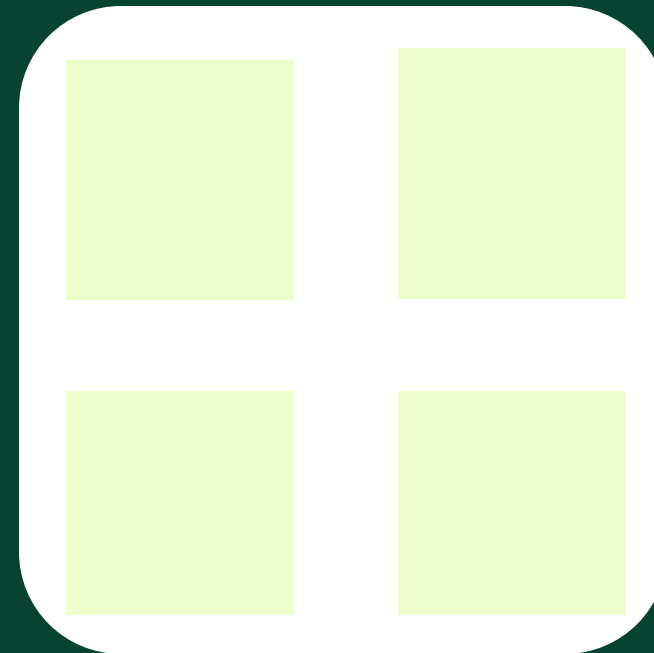
HOW ANOMALYDINO DETECTS THE DEFECT

INPUT IMAGE



DIVIDED INTO 14×14 PATCHES

MEMORY BANK



CALCULATIONS

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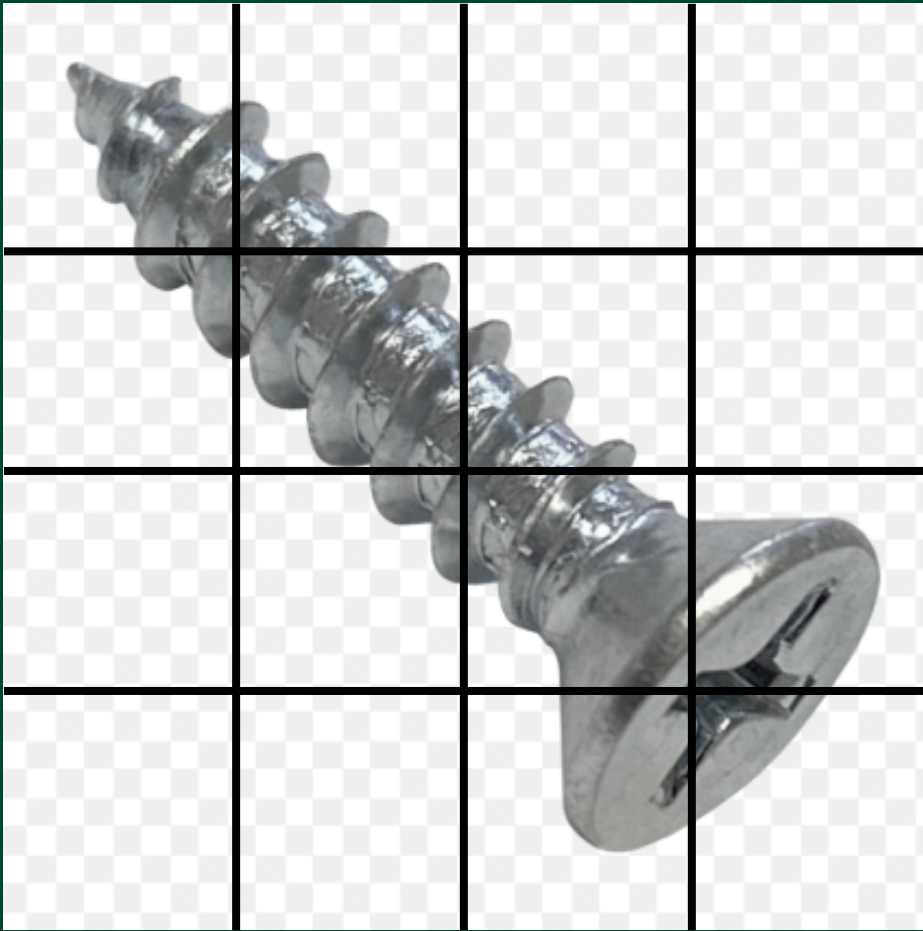
PATCH-LEVEL DISTANCES



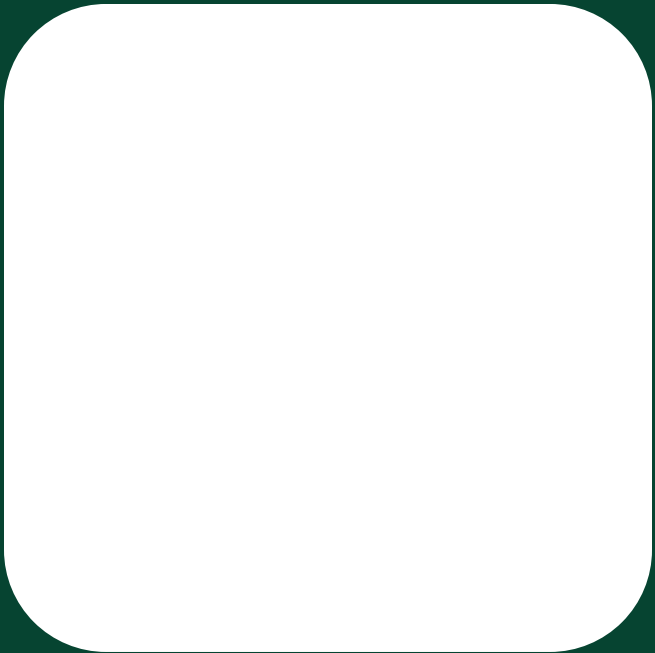
0.82

HOW ANOMALYDINO DETECTS THE DEFECT

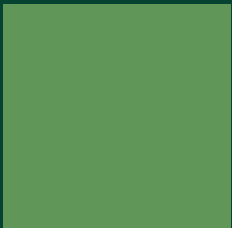
INPUT IMAGE



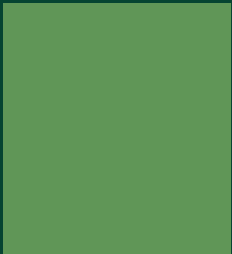
MEMORY BANK



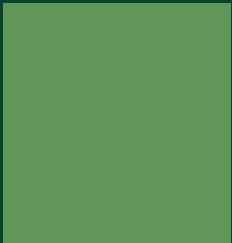
PATCH-LEVEL DISTANCES



0.82



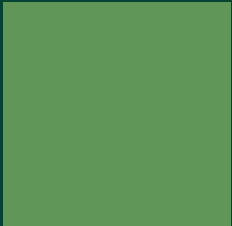
0.74



0.1



0.04



0.001

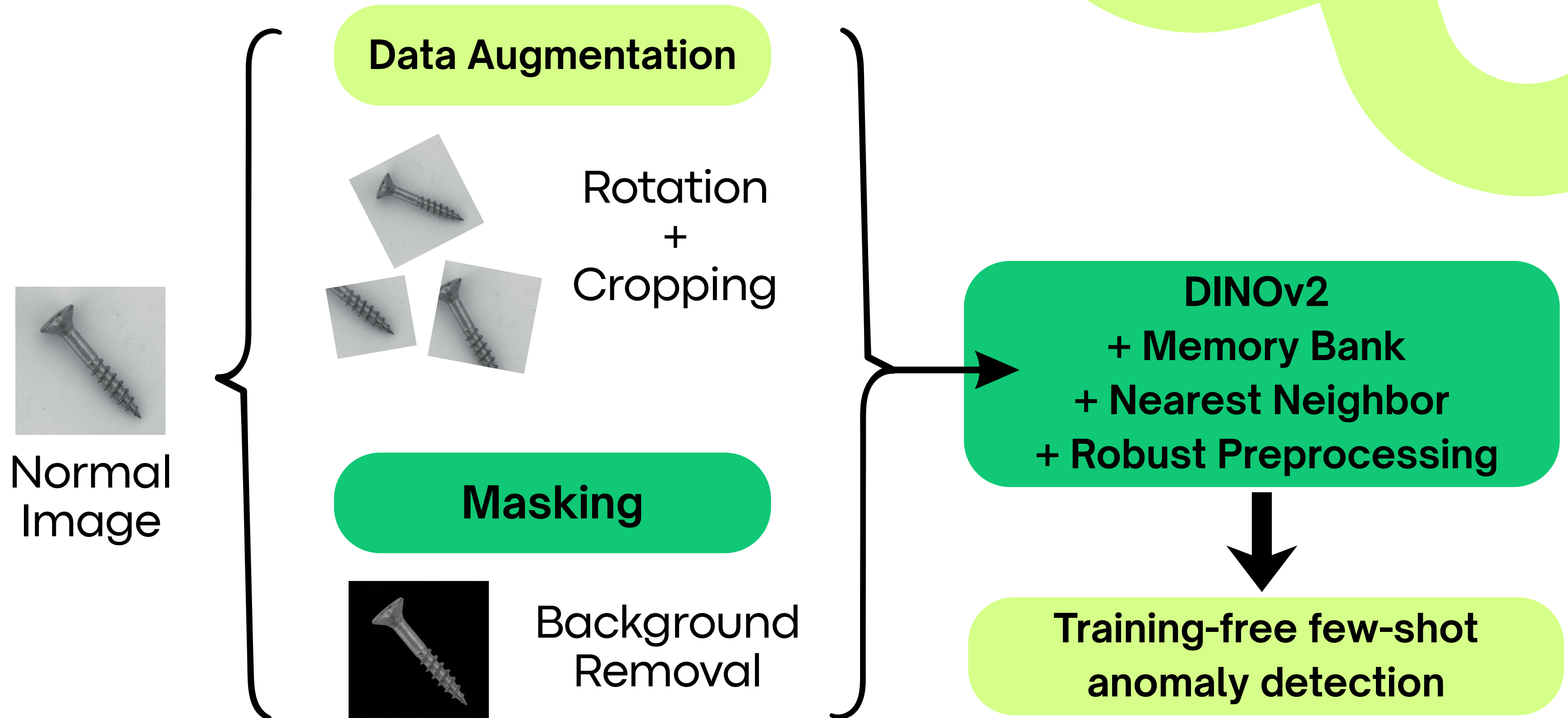
CALCULATIONS

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DIVIDED INTO 14×14 PATCHES

Improving the Memory Bank





Thank You



Team TensorMind